



Project 2 - Web Testing Selenium

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Software Testing and Security Checking

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Selenium

- A suite of tools for automating web browsers
- First launched at 2002/04/30
- Written in Java
- A cross platform software
- Support programming languages
 - C#, Javascript, Java, Python, Ruby





Selenium IDE

- A browser extension that records and plays back a user's actions
- Web ready
- Easy debugging
- Cross-browser execution





Selenium WebDriver

- Designed as a simple and more concise programming interface





Selenium Grid

- Provide an easy way to run tests in parallel on multiple machines
- Allow testing on different browser versions
- Enable cross platform testing



Installation (Selenium IDE)

- Install [Selenium IDE extension](https://chrome.google.com/webstore/detail/selenium-ide/mooikfkahbdckldjjndioackbalphokd) on Chrome/FireFox
 - use Chrome here
 - <https://chrome.google.com/webstore/detail/selenium-ide/mooikfkahbdckldjjndioackbalphokd>



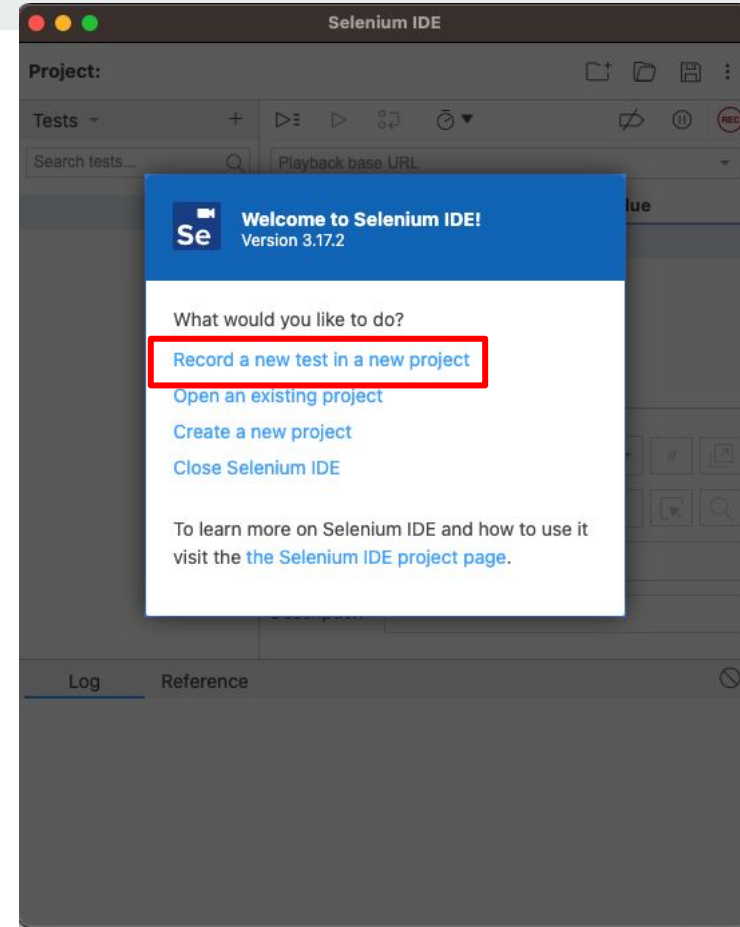


Start with Selenium IDE

1. Create a new project / Open an old project
2. Add a new test to project
3. For a single test in a project
 - a. Recording
 - b. Insert and Delete commands
 - c. Playback
 - d. Export to programming language
4. Save the project into .side file

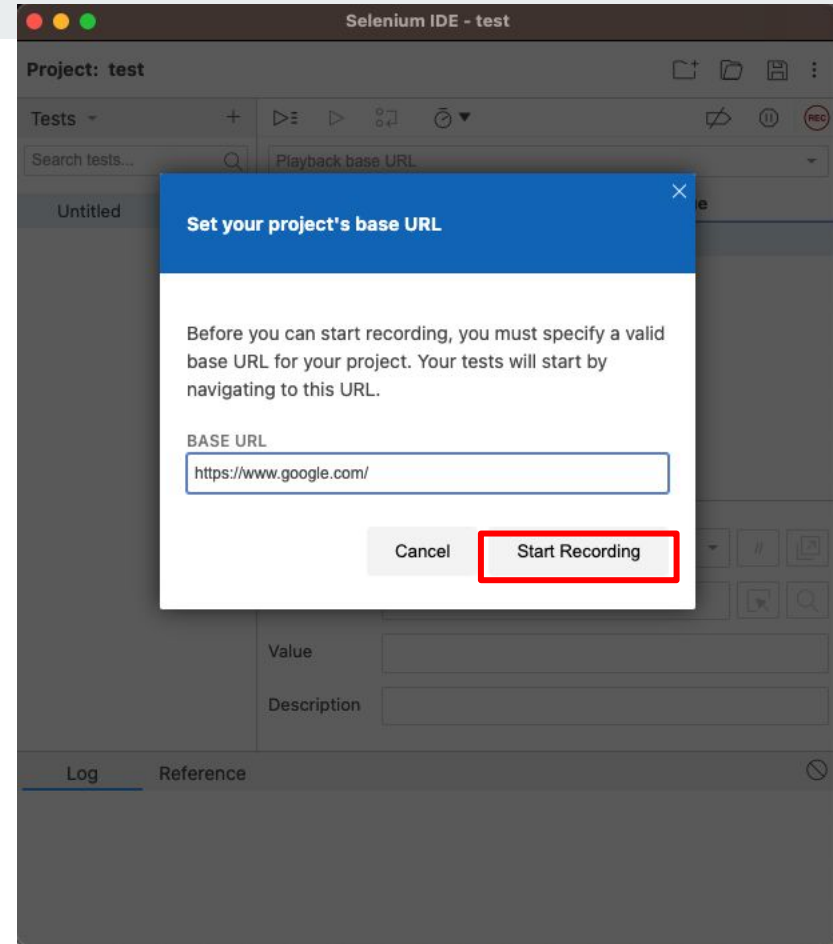
Record a Test

- Record your actions and save as a list of commands
- e.g.
 - open <https://www.google.com/>
 - type 'facebook' in search bar
 - press enter
 - click 'Facebook' page
 - close the page



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☐ Right click on the page

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 - close the page



	Command	Target	Value
1	open	/	
2	set window size	1647x924	
3	type	name=q	facebook
4	send keys	name=q	\${KEY_ENTER}
5	mouse over	css=.tF2Cxc > .yuRUbf .LC20lb	
6	click	css=.tF2Cxc > .yuRUbf .LC20lb	
7	close		

Run a Test

- Run the recorded actions
- Selenium IDE provides result of running with logs

The screenshot displays the Selenium IDE interface for a project named 'fb*'. The interface is divided into several sections:

- Project: fb***: Located at the top left.
- Executing**: A dropdown menu showing the current test suite 'fb*'.
- URL**: A text field containing 'https://www.google.com'.
- Command Table**: A table listing the test steps with columns for Command, Target, and Value.
- Command Form**: A section for defining new commands with fields for Command, Target, Value, and Description.
- Log**: A section showing the execution log of the test.

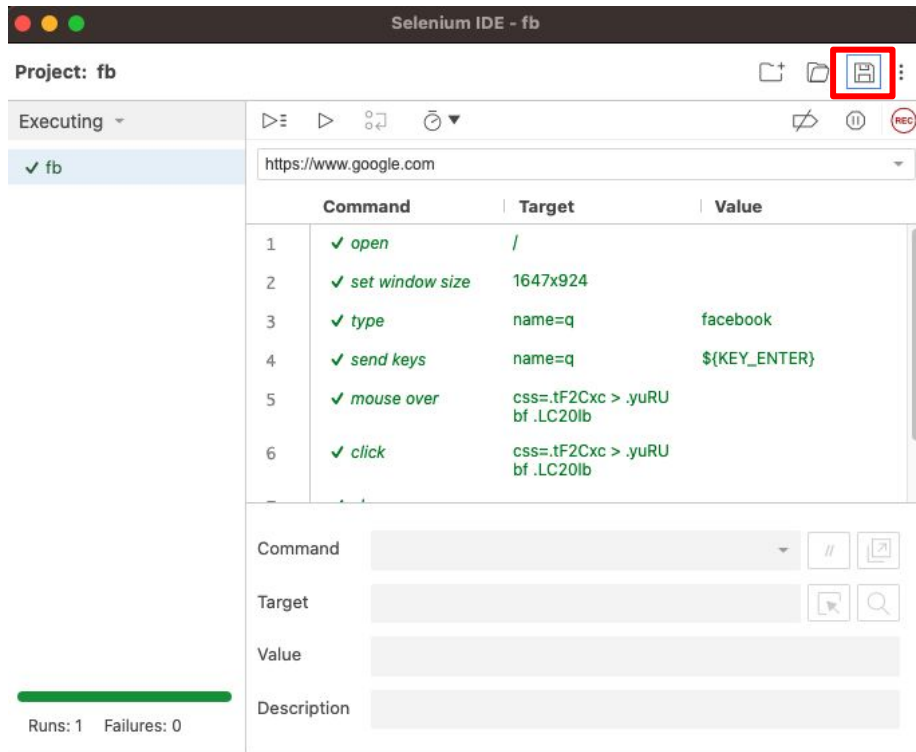
	Command	Target	Value
1	✓ open	/	
2	✓ set window size	1647x924	
3	✓ type	name=q	facebook
4	✓ send keys	name=q	\${KEY_ENTER}
5	✓ mouse over	css=.tF2Cxc > .yuRUbf .LC20lb	
6	✓ click	css=.tF2Cxc > .yuRUbf .LC20lb	
7	✓ close		

Below the table, the Command form is visible with fields for Command, Target, Value, and Description.

The Log section shows the following entries:

```
Running 'fb' 16:25:45
1. open on / OK 16:25:45
2. setWindowSize on 1647x924 OK 16:25:45
3. type on name=q with value facebook OK 16:25:45
4. sendKeys on name=q with value ${KEY_ENTER} OK 16:25:46
5. mouseOver on css=.tF2Cxc > .yuRUbf .LC20lb OK 16:25:46
6. click on css=.tF2Cxc > .yuRUbf .LC20lb OK 16:25:47
7. close OK 16:25:47
'fb' completed successfully 16:25:47
```

- Save the recorded actions to .side file



Export a Test

- Export the recorded actions into programming language

The screenshot displays the Selenium IDE interface for a project named 'fb*'. The 'Tests' panel on the left shows a test named 'fb' with a green checkmark. A right-click context menu is open over the 'fb' test, with the 'Export' option highlighted by a red rectangle. The main panel shows the test steps in a table format:

Command	Target	Value
open	/	
set window size	1647x924	
type	name=q	facebook
send keys	name=q	\${KEY_ENTER}
mouse over	css=.tF2Cxc > .yuRUbf .LC20lb	
click	css=.tF2Cxc > .yuRUbf .LC20lb	
close		

Below the table, the details for the selected 'open' command are shown:

Command: open
Target: /
Value:
Description:

The Log panel at the bottom shows the execution results:

Log	Reference
4. sendKeys on name=q with value \${KEY_ENTER} OK	23:20:27
5. mouseOver on css=.tF2Cxc > .yuRUbf .LC20lb OK	23:20:29
6. click on css=.tF2Cxc > .yuRUbf .LC20lb OK	23:20:30
7. close OK	23:20:31
'fb' completed successfully 23:20:31	



Export a Test

- Export the recorded actions into programming language

Select language

☐ C# NUnit

☐ C# xUnit

☐ Java JUnit

☐ JavaScript Mocha

☒ Python pytest

☐ Ruby RSpec

☐ Include origin tracing code comments

☐ Include step description as a separate comment

☐ Export for use on Selenium Grid

Cancel

Export



Requirement & Evaluation of Project2

- Find one website with multiple language choices
 - Register the website different from other students
 - https://docs.google.com/spreadsheets/d/13_GNUfqFi0Wk2N2mh-DcKhDZwMYpm6iAKFxQoCV_gk/edit?usp=sharing
- (10%) Use assertions to test whether the contents change appropriately according to language selection
 - (9%) The navigation bar
 - (1%) Webpage title
- (5%) Pick two other pages in the same website and test if language selection is properly remembered
- Total 10% + 5% = 15%



Requirement & Evaluation of Project2 (cont.)

- Write a report specifies
 - the webpage you choose
 - test script you design
 - testing result



Submission

- Hand in the following files to NTU COOL
 - your testing script (.side)
 - report
- **Deadline: 2023/5/3 23:59**